

06/01/2020

Asymptotes and Rectangular Hyperbola

An asymptote to a hyperbola is a straight line which meets the hyperbola at infinity at two points but itself do not altogether lie at infinity.

Q. Find the eqn of asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Given hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

Let $y = mx + c$ be an asymptote to parabola.
hyperbola. --- (2)

Then the x coordinates of the point of intersection of (1) and (2) are obtained by eliminating y b/w (1) and (2)

$$\Rightarrow \frac{x^2}{a^2} - \frac{(mx+c)^2}{b^2} = 1$$

$$\Rightarrow b^2 x^2 - a^2 (mx+c)^2 = a^2 b^2$$

$$\Rightarrow x^2 (b^2 - a^2 m^2) - x (2a^2 mc) - a^2 (c^2 + b^2) = 0$$

$$\Rightarrow x^2 (b^2 - a^2 m^2) - x (2a^2 mc) - a^2 (c^2 + b^2) = 0 \quad \text{--- (3)}$$

Now eqn (3) will be an asymptote of (1) if both roots of (3) are infinite.

$\therefore$ coeff. of $x^2 = 0$ & coeff. of $x = 0$

$$\Rightarrow b^2 - a^2 m^2 = 0 \quad \text{and} \quad 2a^2 mc = 0$$

$$\Rightarrow \underline{\underline{m = \pm b/a}}$$

$$\underline{\underline{c = 0}}$$

$$(2) \Rightarrow y = \pm \frac{b}{a} x$$

$\therefore y = \pm \frac{b}{a} x$ are the eqns of asymptotes.

Now $y = \frac{b}{a}x$ & $y = -\frac{b}{a}x$

$$\frac{y}{b} - \frac{x}{a} = 0 \quad \& \quad \frac{y}{b} + \frac{x}{a} = 0$$

Combined eqn of asymptote

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

Q. S.T. the polar of any point on an asymptote w.r.t the hyperbola is $\parallel$ to that asymptote

Let $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be the given hyperbola. Then one of its asymptotes $y = \frac{b}{a}x$.

Let (x_1, y_1) be a point on this asymptote. Then

$$y_1 = \frac{b}{a}x_1 \quad \text{--- (1)} \quad \Rightarrow \quad \frac{x_1}{y_1} = \frac{a}{b}$$

Also polar $(xx_1) - \frac{yy_1}{b^2} = 1$ w.r.t hyperbola

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \quad \text{--- (2)}$$

Slope of (1) $\Rightarrow \frac{b}{a}$

Slope of (2) $\Rightarrow \frac{b^2}{a^2} \left(\frac{x_1}{y_1} \right)$

$$\therefore \frac{x_1}{y_1} = \frac{a}{b}$$

$$\Rightarrow \text{Slope of (2)} \Rightarrow \frac{b}{a}$$

$$\frac{xx_1}{a^2} = \frac{1^2 + y_1^2}{b^2}$$

$$\frac{2xx_1 - a^2}{a^2} = \frac{2y_1}{b^2}$$

$$\frac{b^2(2xx_1 - a^2)}{y_1 a^2} = y_1$$

$\therefore$ Polar of any point on an asymptote w.r.t hyperbola is $\parallel$ to that asymptote.

Date 13/01/20

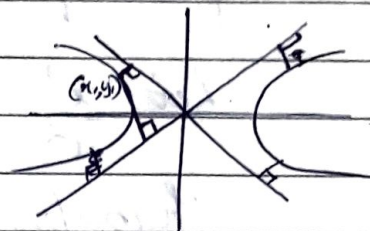
Distance of (x, y) from $Ax + By + C = 0$

$$D = \frac{|Ax + By + C|}{\sqrt{A^2 + B^2}}$$

Q. S.T product of $\perp^r$ from any point of a hyperbola to its asymptotes is constant.

Ean of hyperbola,

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



Let (x_1, y_1) be any point, Then,

$$\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$$

Asymptotes of hyperbola are

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

i.e. $\left(\frac{x+y}{a} + \frac{y-x}{b}\right) \left(\frac{x-y}{a} - \frac{y-x}{b}\right) = 0$

$\therefore$ Length of the $\perp^r$ drawn from (x_1, y_1) to the asymptote are:

$$\frac{\frac{x_1}{a} + \frac{y_1}{b}}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}$$

and $\frac{\frac{x_1}{a} - \frac{y_1}{b}}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}$

$\therefore$ product of $\perp^r$

$$\frac{\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}}{\frac{1}{a^2} + \frac{1}{b^2}} \Rightarrow$$

$$\therefore \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

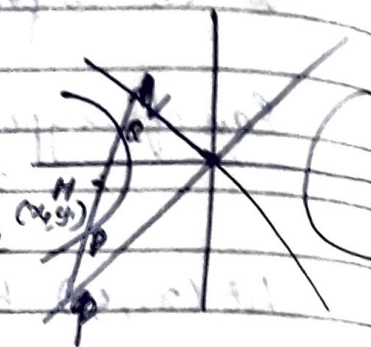
$$\Rightarrow \text{Product} = \frac{1}{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{a^2 b^2}{a^2 + b^2} = k \text{ constant}$$

Hence proved

Imp A chord PQ of a hyperbola intersect its asymptotes p and q. $ST P_p = Q_q$

Let $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be the given

hyperbola and $M(x_1, y_1)$ be the midpoint of PQ.



$\therefore$ eqn of PQ with midpoint M.

$$T_1 = S_1$$

To P.T $P_p = Q_q$
we prove
 $\odot PM = QM$

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} \quad \text{--- (1)}$$

Also asymptotes of hyperbola are

$$\frac{x}{a} + \frac{y}{b} = 0 \quad \text{and} \quad \frac{x}{a} - \frac{y}{b} = 0$$

(2)

(3)

Solving (1) and (2)

Sub $\left(\frac{x}{a} = -\frac{y}{b}\right)$ in (1)

(5)

$$\Rightarrow \frac{-yx_1}{ab} - \frac{yy_1}{b^2} = \left(\frac{x_1}{a} + \frac{y_1}{b}\right) \left(\frac{x_1}{a} - \frac{y_1}{b}\right)$$

$$\Rightarrow \frac{-y}{b} \left(\frac{x_1}{a} + \frac{y_1}{b}\right) = \left(\frac{x_1}{a} + \frac{y_1}{b}\right) \left(\frac{x_1}{a} - \frac{y_1}{b}\right)$$

$$\Rightarrow \underline{y = -b \left(\frac{x_1}{a} - \frac{y_1}{b}\right)} \quad \text{--- (4)}$$

Sub (4) in (5)

$$\frac{x}{a} = -\frac{y}{b} \Rightarrow \underline{\frac{x}{a} = \left(\frac{x_1}{a} - \frac{y_1}{b}\right)}$$

$$\Rightarrow x = a \left(\frac{x_1}{a} - \frac{y_1}{b} \right)$$

$$\therefore p \text{ be } \left(a \left(\frac{x_1}{a} - \frac{y_1}{b} \right), -b \left(\frac{x_1}{a} - \frac{y_1}{b} \right) \right)$$

Solving (1) and (3)

$$\text{Sub } \frac{x}{a} = \frac{y}{b}$$

$$\Rightarrow \frac{y}{b} = \left(\frac{x_1}{a} - \frac{y_1}{b} \right) = \left(\frac{x_1}{a} + \frac{y_1}{b} \right) \left(\frac{x_1 - y_1}{a + b} \right)$$

$$\Rightarrow y = b \left(\frac{x_1}{a} + \frac{y_1}{b} \right)$$

$$\text{iii}^{\text{rd}} \quad x = a \left(\frac{x_1}{a} + \frac{y_1}{b} \right)$$

$$\therefore q \text{ is } \left(a \left(\frac{x_1}{a} + \frac{y_1}{b} \right), b \left(\frac{x_1}{a} + \frac{y_1}{b} \right) \right)$$

Mid point of pq is

$$\left(\frac{a \left(\frac{x_1}{a} - \frac{y_1}{b} \right) + a \left(\frac{x_1}{a} + \frac{y_1}{b} \right)}{2}, \frac{-b \left(\frac{x_1}{a} - \frac{y_1}{b} \right) + b \left(\frac{x_1}{a} + \frac{y_1}{b} \right)}{2} \right)$$

$$\Rightarrow \left(\frac{x_1 - \frac{ay_1}{b} + x_1 + \frac{ay_1}{b}}{2}, \frac{-\frac{bx_1}{a} + y_1 + \frac{bx_1}{a} + y_1}{2} \right)$$

$$\Rightarrow \left(\frac{2x_1}{2}, \frac{2y_1}{2} \right) \Rightarrow (x_1, y_1)$$

$\therefore$ Mid point of PR = Mid point of pq

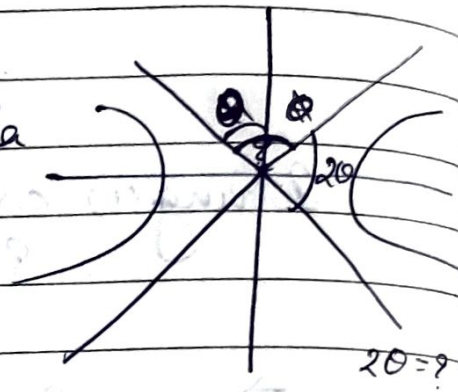
$$\therefore P_R = Q_R$$

Q. Find the angle b/w the asymptotes of a hyperbola.

The eqn of asymptotes of a hyperbola are given by

$$0 = \left(\frac{x}{a} + \frac{y}{b}\right) \left(\frac{x}{a} - \frac{y}{b}\right) = 0$$

$$\Rightarrow y = \frac{b}{a}x \text{ and } -\frac{b}{a}x$$



They are equally inclined to the x -axis if 2θ is the angle b/w asymptotes.

$$\Rightarrow \tan \theta = \frac{b}{a}$$

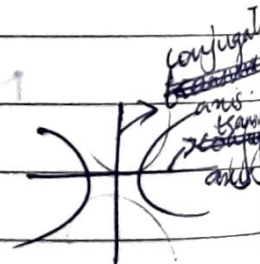
$$\therefore y = \frac{b}{a}x$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$= 1 + \frac{b^2}{a^2} = \frac{a^2 + b^2}{a^2} = e^2$$

$$\Rightarrow \sec \theta = e$$

$$\Rightarrow \theta = \sec^{-1}(e)$$



$$\therefore \text{Angle b/w asymptotes} = \underline{2 \sec^{-1}(e)}$$